Two clocks: operator-level pressure–temperature decoupling measured across scales

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Abstract

Pressure and temperature in a fluid are driven by the same source but evolve under different operators: temperature obeys a local parabolic advection–diffusion equation, while pressure in the incompressible limit obeys a global elliptic constraint. We show that this operator split is measurable as distinct temporal and spatial signatures across five independent settings. In field data, temperature tracks the local diurnal cycle while pressure carries a planetary-scale semidiurnal tide whose amplitude follows latitude, not local heating. In Rayleigh–Bénard direct numerical simulation, pressure and buoyancy share the same integral scale but differ by a factor of 2.4 in Taylor microscale and $\sim 10^4$ in high-wavenumber spectral power, with the gap widening as Ra increases from 10^6 to 10^8. In controlled compressible solvers, the hyperbolic-to- elliptic crossover occurs on a timescale $t_{90} \approx 2.3\, L/c$, and the fast-clock energy fraction scales as M^2. In the projection method, a single elliptic solve reduces the divergence of the intermediate velocity field from 2.7×10^{-2} to machine zero, while a spectrum-matched phase-randomized surrogate leaves it unrepaired. In NCEP/NCAR reanalysis, the divergent (fast) component of the wind carries $\leq 4\%$ of the kinetic energy, with a minimum at the classical level of non-divergence. These measurements establish that the two-operator structure is a robust, measurable property of real flows, with direct implications for single-diffusivity turbulence closures.

1 Introduction

A fluid heated from below or beside responds in its temperature field and in its pressure field. These two responses are coupled through the equation of state and the momentum balance, yet they are governed by different mathematical operators with different Green's functions, different characteristic times, and different spatial footprints. Temperature is transported: it is advected, diffused, and filamented by the flow. Pressure, in the incompressible limit, is not transported at all; it is the Lagrange multiplier of the divergence-free constraint, determined instantaneously by a global elliptic solve. At finite compressibility, that instantaneous adjustment is replaced by acoustic waves propagating at speed c, but the adjustment remains fast compared to the advective timescale whenever the Mach number is small.

This distinction is elementary at the level of the governing equations. Its consequences for turbulence modeling, however, are rarely stated explicitly. The standard closure assumption — a single scalar eddy diffusivity acting on resolved gradients, optionally with a fixed turbulent Prandtl number (Boussinesq 1877; Smagorinsky 1963) — treats all transported quantities as if they respond to stirring in the same way, on the same timescale, with the same spatial structure. If pressure and temperature genuinely evolve under different operators, this assumption is not merely approximate; it is structurally incomplete.

The purpose of this paper is to establish, quantitatively and across multiple independent settings, that the operator split is measurable. We do not propose a new closure; that is the subject of a companion paper (Saifelden 2026, hereafter **P1**). We do not claim that single-diffusivity models are useless; we claim that they are one-clock models applied to a two-clock system, and we measure the discrepancy.

The paper proceeds as follows. Section 2 states the two-operator structure and identifies the relevant timescales. Section 3 measures the temporal decoupling in atmospheric field data. Section 4 measures the spatial decoupling in direct numerical simulation. Section 5 demonstrates the mechanism in controlled compressible and incompressible solvers. Section 6 confirms the energy split in the real atmospheric wind field. Section 7 discusses implications and limitations.

2 The two operators

2.1 Governing equations

Let $\mathbf{u}$ be velocity, p pressure, θ a transported scalar (temperature or buoyancy), and ν, κ the molecular viscosity and diffusivity. The scalar obeys the parabolic advection–diffusion equation

$$\partial_t \theta + (\mathbf{u} \cdot \nabla)\theta = \kappa \nabla^2 \theta + S_\theta, \tag{1}$$

whose Green's function in the diffusive limit is a spreading Gaussian of width $\sqrt{4\kappa t}$: local, causal, and memory-bearing.

In an incompressible flow, pressure is determined by the elliptic Poisson equation obtained from the divergence of the momentum equation:

$$\nabla^2 p = -\rho \nabla \cdot [(\mathbf{u} \cdot \nabla)\mathbf{u}] + \nabla \cdot \mathbf{f}. \tag{2}$$

Its Green's function is $\sim \ln r$ in two dimensions ($\sim -1/r$ in three dimensions): global, instantaneous, and boundary-aware. In Fourier space, $\hat{p} = -\hat{f}/k^2$; division by k^2 is a low-pass filter that suppresses small-scale power by a factor $\sim 1/k^4$ relative to the source.

At finite compressibility, pressure instead satisfies the hyperbolic wave equation

$$\frac{1}{c^2}\,\partial_{tt} p - \nabla^2 p = s(\mathbf{x}, t), \tag{3}$$

with sound speed c. As $M = U/c \to 0$, Eq. (3) degenerates into Eq. (2): the waves become infinitely fast and vanish into the constraint (Klainerman & Majda 1981).

2.2 Timescales

Three timescales are relevant for a domain of size L, flow speed U, and eddy scale ℓ:

$$\tau_p \sim \frac{L}{c}, \qquad \tau_{\text{adv}} \sim \frac{\ell}{U}, \qquad \tau_\kappa \sim \frac{\ell^2}{\kappa}. \tag{4}$$

For air at laboratory-to-mesoscale L, $\tau_p/\tau_{\text{adv}} = M(L/\ell)^{-1} \ll 1$. Pressure equilibrates before the flow moves; scalars integrate their history as the flow evolves. We refer to the parabolic timescale as the *slow clock* and the elliptic/acoustic timescale as the *fast clock*.

3 Temporal decoupling in field data

We first ask whether the two clocks are distinguishable in time-series measurements at fixed locations. If pressure and temperature respond to the same forcing on the same timescale, their spectra should be similar and their bulk correlation should be tight. If they respond through different operators, their spectra should differ qualitatively and their correlation should be weak and regime-dependent.

3.1 Single-site measurement: NEON eddy covariance

We use the NEON bundled eddy-covariance product (DP4.00200.001) at the Wind River Experimental Forest site (WREF; 45.82°N, 121.95°W; 53 m canopy height) for January 2020 at 30-minute resolution: 1488 intervals, of which 1097 carry valid pressure and 1180 valid temperature (NEON 2020).

Figure 1 shows diurnal composites of incoming shortwave radiation, air temperature, sensible heat flux, and barometric pressure. Three features are apparent.

First, the source–response chain is causal and ordered: incoming shortwave leads air temperature by approximately two hours and the sensible heat flux by approximately three hours. The temperature signal is driven locally by solar heating.

Second, the spectral characters of the two fields are qualitatively different. Temperature variance is concentrated at the 24-hour diurnal line. Pressure variance is dominated by multi-day synoptic variability with a distinct secondary peak at the 12-hour semidiurnal atmospheric tide, visible as twin daily maxima in the raw composite.

Third, the bulk daily correlation between pressure and temperature is weak: $r(P, T) \approx +0.07$ ($R^2 \approx 0.005$). This is far from the tight positive coupling that a constant-density ideal-gas relation would predict. The mismatch is accommodated by density fluctuations of order one percent; the gas law holds locally, but bulk pressure does not track bulk temperature.

A seasonal cross-check using July 2020 at the same site reproduces the qualitative fingerprint at the opposite solar extreme. The temperature diurnal signal becomes even cleaner (semidiurnal-to-diurnal power ratio drops from 7.3% in winter to 1.0% in summer), pressure retains its semidiurnal tide component, and the bulk daily correlation shifts to $r = -0.49$ — the summer thermal-low regime, in which hot afternoons lower surface pressure. The sign and magnitude of the bulk correlation are set by the synoptic regime; the spectral separation between the two fields is not.

3.2 Multi-station measurement: the latitude fingerprint of a global mode

A single tower cannot distinguish "pressure is noisy" from "pressure responds to the whole atmosphere." To make this distinction, we examine one-minute barometric and thermometric data from eight ASOS stations spanning 24.6°–47.4°N (Iowa Environmental Mesonet archive; Q1 2020).

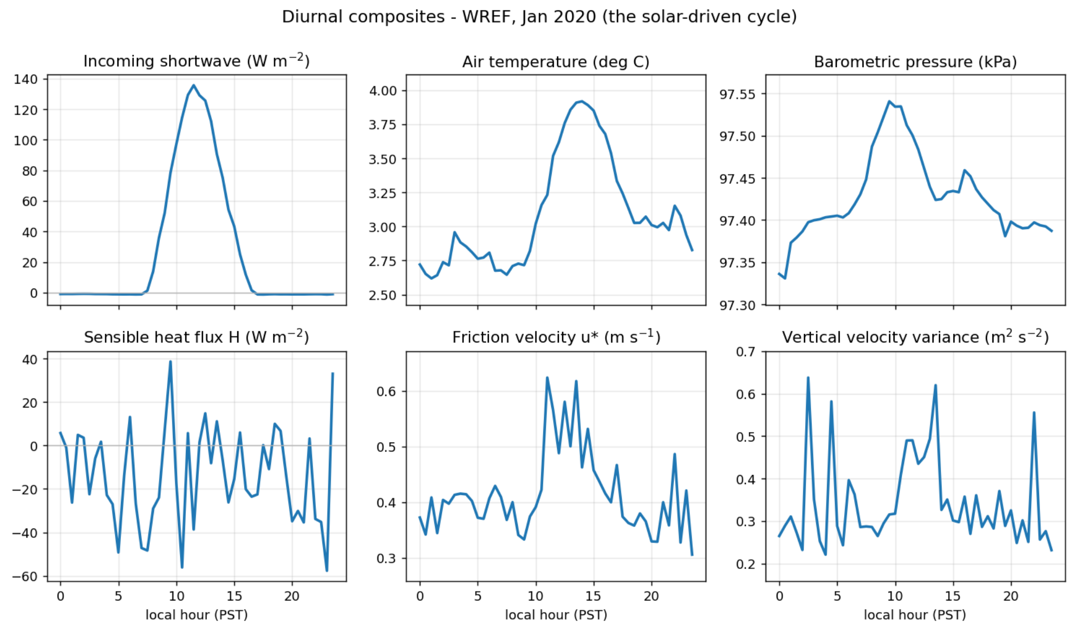

Figure 1: NEON WREF, January 2020: diurnal composites of incoming shortwave radiation, air temperature, sensible heat flux, and barometric pressure. Temperature exhibits a single clean 24-hour signal driven by local solar heating. Pressure shows a twin-peak semidiurnal tide superimposed on multi-day synoptic variability.

Confidence intervals are computed via daily moving-block bootstrap ($n = 500$).

Table 1 summarizes the key measurements. The 24-hour temperature amplitude varies with local insolation and surface type: it is largest at the interior desert site (PHX, 5.06°C) and smallest at the marine sites (SEA, 1.79°C; EYW, 1.51°C). It does not order monotonically with latitude.

The 12-hour pressure-tide amplitude, by contrast, decreases monotonically from 1.10 hPa at 24.6°N to 0.34 hPa at 47.4°N. This is the classical signature of the global solar semidiurnal tide — a planetary normal-mode response whose amplitude is largest where the forcing projects onto the low-latitude waveguide (Chapman & Lindzen 1970). The trend correlation with the migrating-tide climatology $A_H = 1.16 \cos^3 \varphi$ hPa (Haurwitz 1956) is 0.98. Station totals fall in the range 0.9–$1.4 \times A_H$, consistent with the expected contribution of non-migrating components (Dai & Wang 1999).

The interpretation is direct: pressure at a fixed site carries the signature of a planetary-scale resonance, while temperature carries the signature of the local solar cycle. The two fields share a forcing (the sun) but respond through different operators with different spatial footprints. Station-level daily pressure–temperature correlations range from −0.05 to −0.56, with bootstrap 95% confidence intervals excluding any tight positive lock: the bulk coupling is set by the synoptic regime, not by a gas-law link between the two fields.

Band-resolved coherence sharpens this statement. The magnitude-squared coherence between hourly pressure and temperature is high (0.82–0.98) at the solar harmonic lines, where both fields are phase-locked to the same astronomical forcing, and above the phase-randomized surrogate 95% level at all eight stations. In the off-line sub-diurnal band (9–11 h and 13–20 h), coherence collapses to the surrogate noise floor. Away from the shared solar clock, nothing locks the two fields together.

Table 1: Semidiurnal pressure-tide amplitude and diurnal temperature amplitude at eight ASOS stations, Q1 2020. Bootstrap 95% confidence intervals ($n = 500$) are given for the pressure-tide amplitude. The ratio to the Haurwitz migrating-tide climatology $A_H = 1.16 \cos^3 \varphi$ is shown in the final column.

Station	φ (°N)	$S_{1,T}$ (°C)	$S_{2,P}$ [95% CI] (hPa)	$S_{2,P}/A_H$
EYW	24.6	1.51	1.103 [1.013, 1.191]	1.26
MIA	25.8	2.73	1.077 [0.979, 1.169]	1.27
MSY	30.0	2.75	1.018 [0.878, 1.169]	1.35
DFW	32.9	3.28	0.957 [0.794, 1.141]	1.39
PHX	33.4	5.06	0.924 [0.861, 1.008]	1.37
DSM	41.5	2.73	0.628 [0.398, 0.965]	1.29
CAR	46.9	3.03	0.439 [0.181, 0.879]	1.19
SEA	47.4	1.79	0.337 [0.262, 0.529]	0.94

4 Spatial decoupling in direct numerical simulation

Field sensors measure time series at a point; they cannot resolve spatial structure. A direct numerical simulation can. We use the two-dimensional Rayleigh–Bénard configuration from The Well dataset (Ohana et al. 2024): $Ra = 10^6$, $Pr = 1$, 512×128 grid, periodic in the horizontal, no-slip walls, with velocity, buoyancy, and pressure co-resolved on the same grid.

4.1 Same forcing, two geometries

The central question is whether pressure and buoyancy, driven by the same convective instability, develop the same spatial structure. The answer is no, and the difference is precisely what the operator analysis of Section 2 predicts.

At large scales, the two fields agree: both are organized by the convection rolls, and their integral length scales are nearly identical (0.195 for buoyancy, 0.207 for pressure; ratio 1.06). At small scales, they diverge sharply. The Taylor microscale of pressure is 2.40 times that of buoyancy, and the high-wavenumber spectral power of buoyancy exceeds that of pressure by a factor of approximately 10^4 at $k \approx 10$.

Both numbers are direct consequences of the inverse Laplacian in Eq. (2). The factor $1/k^2$ in Fourier space suppresses pressure power by $\sim 1/k^4$ relative to its source, so pressure remains smooth and domain-spanning while buoyancy is advected into thin filaments with intense small-scale gradients. Figure 3 shows a snapshot; Figure 4 shows the spectra.

4.2 Robustness across Rayleigh number

The microscale ratio is not an artifact of a single Rayleigh number. Sweeping $Ra = 10^6$, 10^7, and 10^8 using The Well's $Pr = 1$ test files (5 trajectories $\times$ 6 statistically steady snapshots each) yields the results in Table 2. The Taylor-microscale ratio grows from 2.58 at $Ra = 10^6$ to 4.45 at $Ra = 10^8$, while the integral-scale ratio remains order unity. As Ra increases and buoyancy plumes

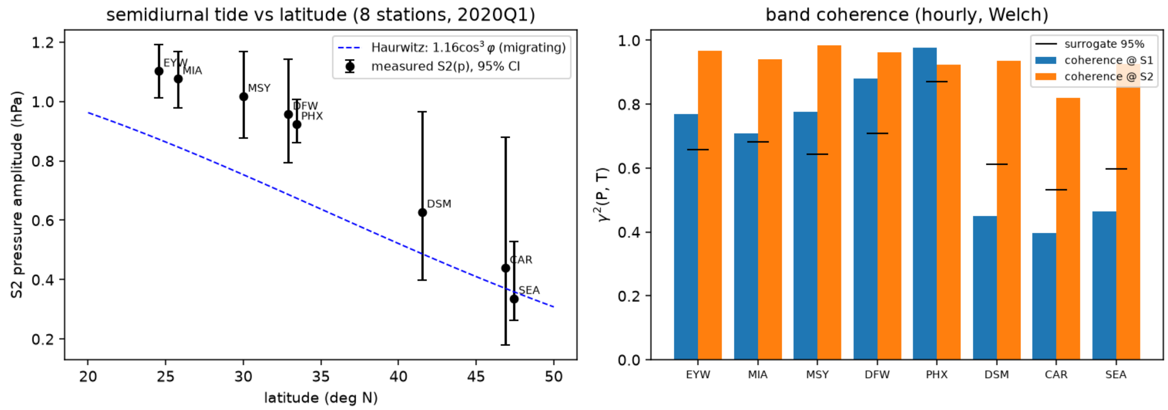

Figure 2: Left: semidiurnal pressure-tide amplitude versus latitude at eight ASOS stations with bootstrap 95% confidence intervals, compared to the Haurwitz migrating-tide climatology (dashed curve). The monotonic equator-to-pole decay is the signature of a global resonance. Right: band-resolved coherence between hourly pressure and temperature — high at the shared solar lines, at the surrogate noise floor off-line.

Table 2: Taylor-microscale ratio (λ_p/λ_θ) and integral-scale ratio (L_p/L_θ) as functions of Ra, from The Well $Pr = 1$ Rayleigh–Bénard dataset. Values are medians with 5–95% ranges over 30 snapshots per Ra (artifact `15b_rb_ra_sweep.json`).

Ra	λ_p/λ_θ	L_p/L_θ
10^6	2.58 [1.98–3.59]	1.02 [0.88–1.40]
10^7	2.94 [2.54–3.65]	1.25 [1.03–1.85]
10^8	4.45 [3.27–5.88]	1.75 [1.42–2.11]

sharpen toward finer scales, the inverse Laplacian continues to hold pressure at the large scales. The operator split widens with turbulence intensity rather than closing.

4.3 Green's-function illustration

To isolate the operator contrast from turbulence statistics, we apply a single localized source to both the elliptic and parabolic operators in a developed cavity flow. The elliptic response spreads over the entire domain: 79% of its magnitude lies beyond one source-width from the origin. The parabolic response at matched time remains a confined Gaussian blob. Boundary geometry edits the global pressure field instantly and only the local temperature field (Fig. 5).

5 Dynamical demonstrations

The field data and DNS establish that the two-operator structure is measurable. We now demonstrate the mechanism in controlled settings where every parameter is known.

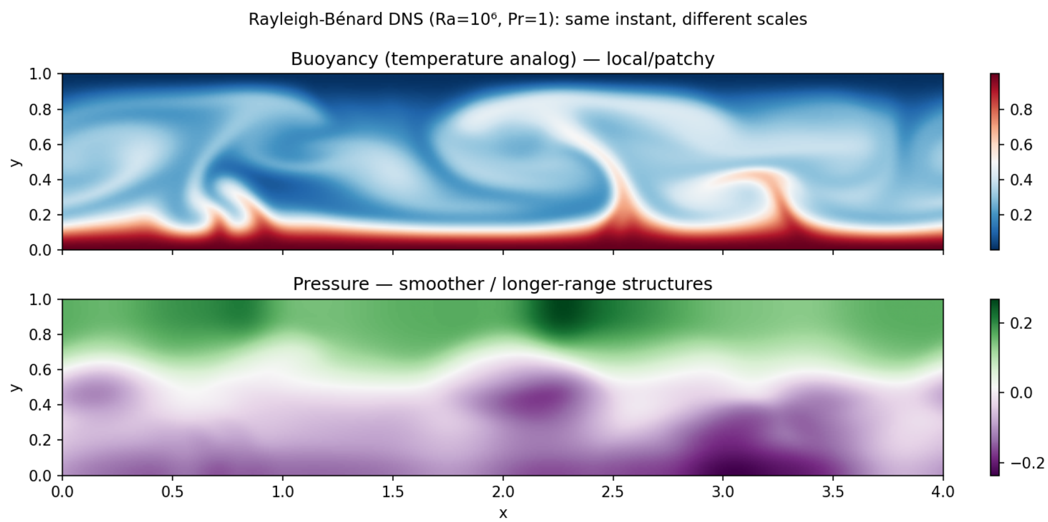

Figure 3: Rayleigh–Bénard DNS snapshot at $Ra = 10^6$: buoyancy (left) is filamentary and sharp, organized into thin plumes; pressure (right) is smooth and domain-spanning. Both fields are from the same simulation at the same instant.

5.1 The hyperbolic-to-elliptic crossover

A minimal two-dimensional linear-acoustics solver makes the relationship between Eqs. (2) and (3) concrete. An initial pressure perturbation radiates as a ring at speed c and decays by geometric spreading. Under sustained forcing, the compressible pressure relaxes to the same field as the elliptic Poisson solve, with a final relative error of $\sim 2 \times 10^{-3}$ for every sound speed tested.

The relaxation time scales as

$$t_{90} \approx 2.3\,\frac{L}{c}, \tag{5}$$

and the error histories for all sound speeds collapse onto a single curve when time is rescaled as ct/L (Fig. 6). The sound speed sets the clock; the final state is independent of it. As $c \to \infty$, the adjustment becomes instantaneous and the hyperbolic equation degenerates into the elliptic constraint.

5.2 Both clocks in one nonlinear flow

A fully nonlinear two-dimensional isothermal compressible Navier–Stokes solver (pseudo-spectral, periodic, with $p = c^2\rho$ so that c is an explicit parameter) is initialized from a Taylor–Green vortex at $Re \approx 600$. The velocity field is Helmholtz-decomposed into solenoidal (vortical, slow) and dilatational (acoustic, fast) components at each time step.

Table 3 summarizes the Mach-number sweep. The dilatational energy fraction scales as $\sim M^2$, vanishing in the incompressible limit. The time-averaged pressure (averaged over the fast acoustic period) converges to the elliptic Poisson pressure with correlation approaching unity and residual shrinking as M^2.

Two clocks coexist in one flow: the vortical energy decays on the slow advective timescale while the acoustic energy oscillates on the fast period (Fig. 7). Averaging over the fast clock acts as a

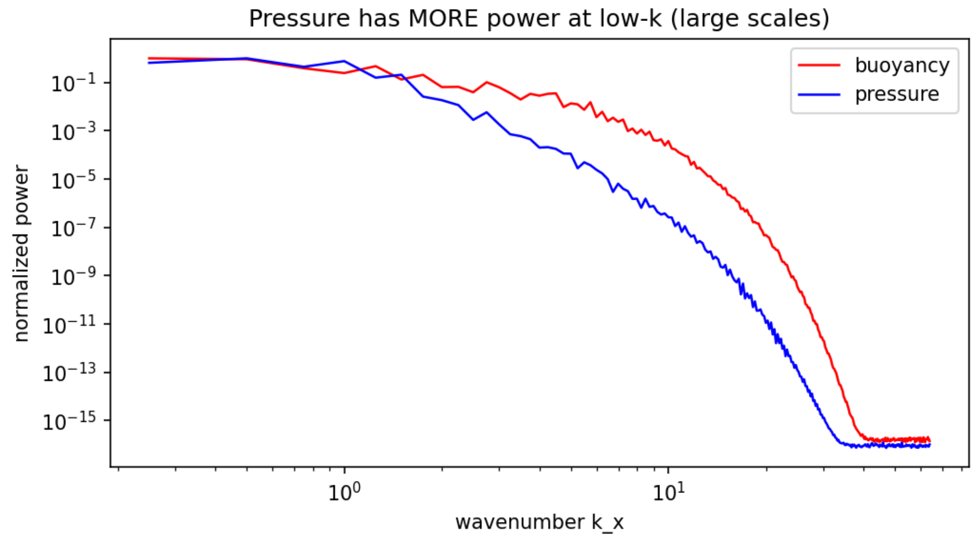

Figure 4: Spatial spectra of buoyancy and pressure from the Rayleigh–Bénard DNS. The energy-containing scales coincide; at high wavenumber, buoyancy exceeds pressure by $\sim 10^4$, consistent with the $1/k^4$ suppression of the inverse Laplacian.

Table 3: Mach-number sweep in the nonlinear compressible solver. $\langle KE_{\text{dil}} \rangle / \langle KE_{\text{sol}} \rangle$ is the time-averaged ratio of dilatational to solenoidal kinetic energy. The pressure residual is the RMS difference between the acoustic-averaged compressible pressure and the elliptic Poisson pressure, normalized by the RMS pressure.

M	$KE_{\text{dil}}/KE_{\text{sol}}$	Pressure residual
0.05	1.5×10^{-4}	8.1×10^{-4}
0.10	5.9×10^{-4}	3.2×10^{-3}
0.20	2.3×10^{-3}	1.4×10^{-2}
0.40	9.3×10^{-3}	1.0×10^{-1}

low-pass filter that recovers the elliptic limit — an operation we will repeat on the real atmosphere in Section 6.

5.3 The projection step as two-clocks synthesis

The standard projection (fractional-step) method advances a Boussinesq flow by a local drift followed by a global elliptic correction (Chorin 1968; Leray 1934):

$$\mathbf{u}^* = \mathbf{u}^n + \Delta t \left[-(\mathbf{u} \cdot \nabla)\mathbf{u} + \nu \nabla^2 \mathbf{u} + (b - \langle b \rangle)\hat{\mathbf{y}} \right], \tag{6}$$

$$\mathbf{u}^{n+1} = \mathbb{P}\, \mathbf{u}^*, \tag{7}$$

$$b^{n+1} = b^n + \Delta t \left[-(\mathbf{u} \cdot \nabla) b + \kappa \nabla^2 b \right], \tag{8}$$

where $\mathbb{P} = I - \nabla(\nabla^2)^{-1}\nabla\cdot$ is the Leray projector. The drift step (Eq. 6) is a local, Markovian update — the parabolic clock. The projection step (Eq. 7) is a single global Poisson solve — the elliptic clock acting as a constraint.

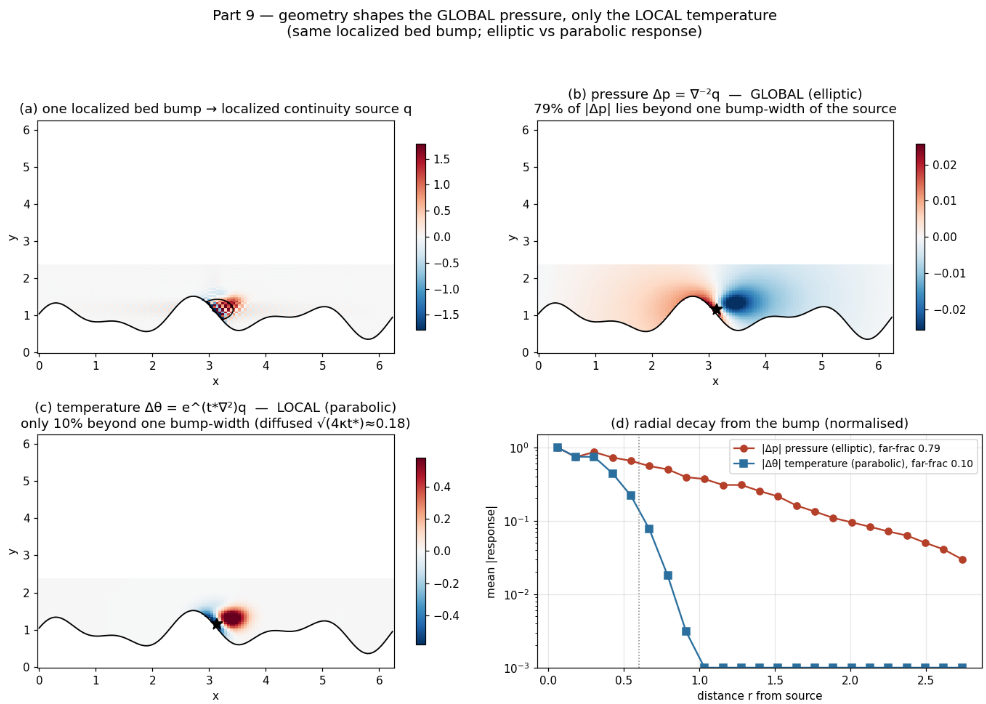

Figure 5: Response to an identical localized source applied to both operators. The elliptic (pressure) response spreads globally, with 79% of its magnitude beyond one source-width. The parabolic (heat-kernel) response remains confined.

Three results follow.

First, the intermediate velocity $\mathbf{u}^*$ violates mass conservation at RMS $\nabla \cdot \mathbf{u}^* \approx 2.7 \times 10^{-2}$. A single projection reduces this to $\approx 2 \times 10^{-15}$ — machine zero (Fig. 9).

Second, replacing the projection correction by a phase-randomized field with the identical power spectrum (Parseval-exact) — the structural move of spectrum-matched stochastic closures (Hasselmann 1976; Berner et al. 2017) — yields a pointwise correlation with the true correction of median $|r| = 0.07$ (95th percentile 0.20, over an ensemble of $n = 100$ surrogates). The RMS divergence after the surrogate correction is 3.7×10^{-2} [3.5–3.8×10^{-2}, 95% interval] — no better than applying no correction at all (2.7×10^{-2}).

Third, the physical interpretation is that the fast clock's action on the slow dynamics is a global elliptic correction with a definite geometric structure, not a stochastic variance to be re-injected with the correct energy. Matching a power spectrum does not reconstruct a boundary-aware, globally coupled elliptic operator.

6 The two clocks in the real atmospheric wind field

The preceding sections establish the operator split in controlled and semi-controlled settings. We now ask whether the same structure is visible in the real atmosphere.

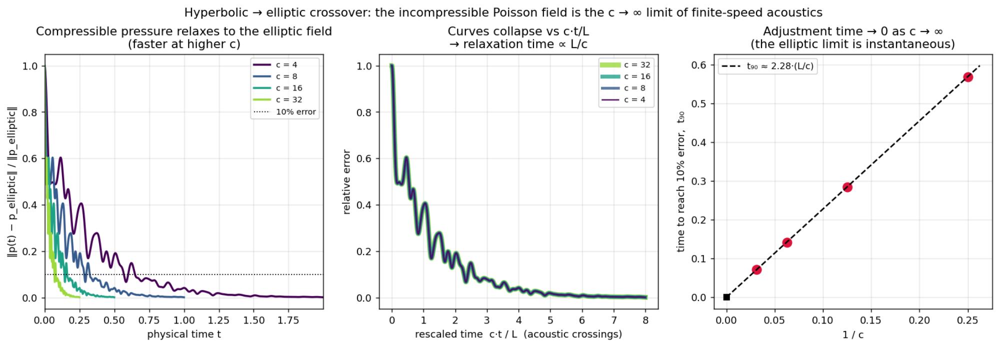

Figure 6: Relaxation of the compressible pressure field to the elliptic (Poisson) solution for several sound speeds. All curves collapse in the rescaled time ct/L, with $t_{90} \approx 2.3\, L/c$.

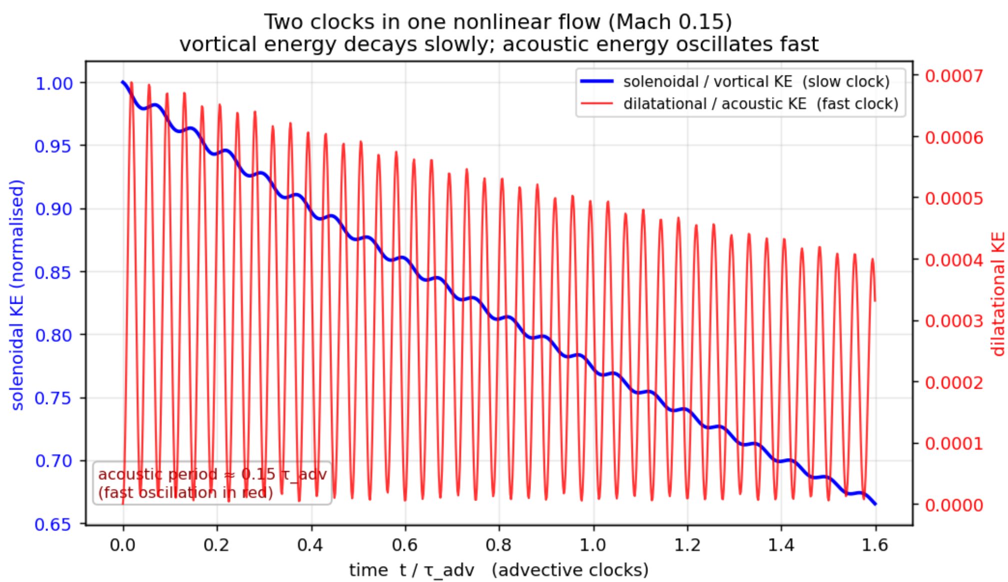

Figure 7: Nonlinear compressible box at $M = 0.1$: slow vortical energy decay with fast acoustic oscillation superimposed. Both clocks are present in one flow.

6.1 Method

Horizontal winds from the NCEP/NCAR Reanalysis 1 (2.5° resolution; Kalnay et al. 1996) are Helmholtz-decomposed on the sphere via spherical-harmonic inversions $\nabla^2\psi = \zeta$ and $\nabla^2\chi = \delta$ into a rotational (non-divergent, balanced) component and a divergent (fast) component. The same decomposition is applied to the NCEP–DOE Reanalysis 2 (Kanamitsu et al. 2002) as an independent cross-check.

Horizontal divergence is not a model artifact: by three-dimensional continuity, it is the genuine footprint of vertical motion. The diagnostic is therefore the energy split, not the existence of divergence.

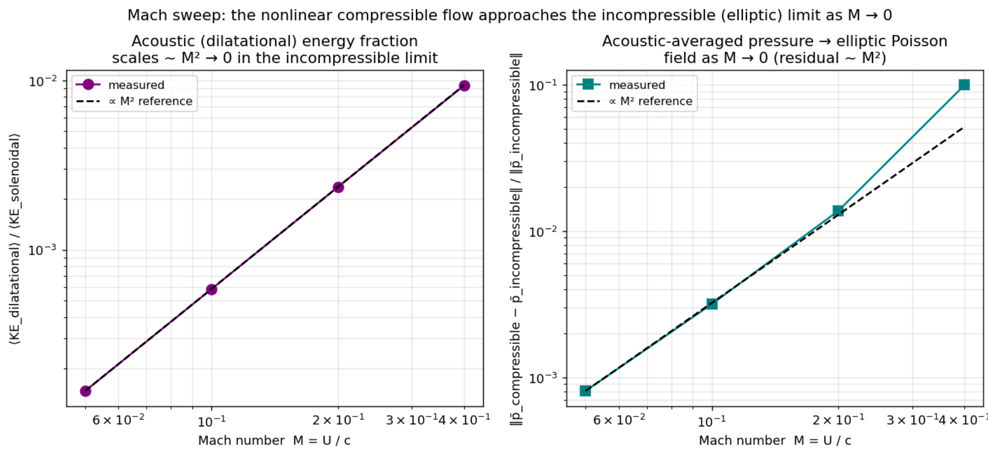

Figure 8: Mach-number sweep: dilatational energy fraction (circles) and acoustic-averaged pressure residual (squares) both scale as $\sim M^2$ (dashed line).

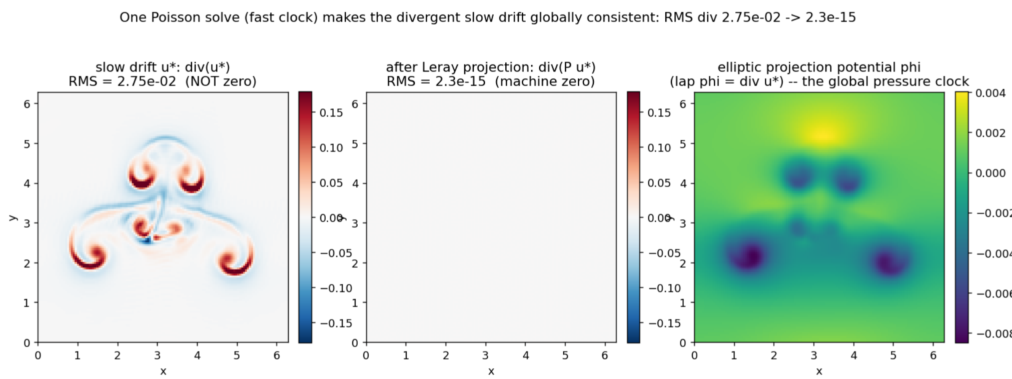

Figure 9: One projection time step, dissected. Top: the divergent intermediate velocity $\mathbf{u}^*$ (RMS $\nabla \cdot \mathbf{u}^* = 2.7 \times 10^{-2}$). Middle: the global elliptic potential ϕ satisfying $\nabla^2\phi = \nabla \cdot \mathbf{u}^*$. Bottom: the projected, solenoidal velocity $\mathbf{u}^{n+1}$ (RMS $\nabla \cdot \mathbf{u}^{n+1} = 2 \times 10^{-15}$).

6.2 Results

Table 4 gives the ratio of divergent to rotational kinetic energy at three pressure levels, from both reanalyses.

Four features emerge.

First, the rotational wind dominates the kinetic energy at every level and every resolved scale. Splitting by spherical-harmonic degree l, the divergent branch sits one to two decades below the rotational branch for all $l \leq 35$ (Fig. 12).

Second, the vertical structure is physically meaningful: the divergent fraction is smallest near 500 hPa — the classical level of non-divergence (Holton 2004) — and larger in the boundary layer and near the jet outflow (Fig. 13).

Third, the cross-check with Reanalysis 2 reproduces the same ordering and the same 500-hPa minimum, with relative differences of 15% or less. The ratio is a property of the analyzed atmosphere,

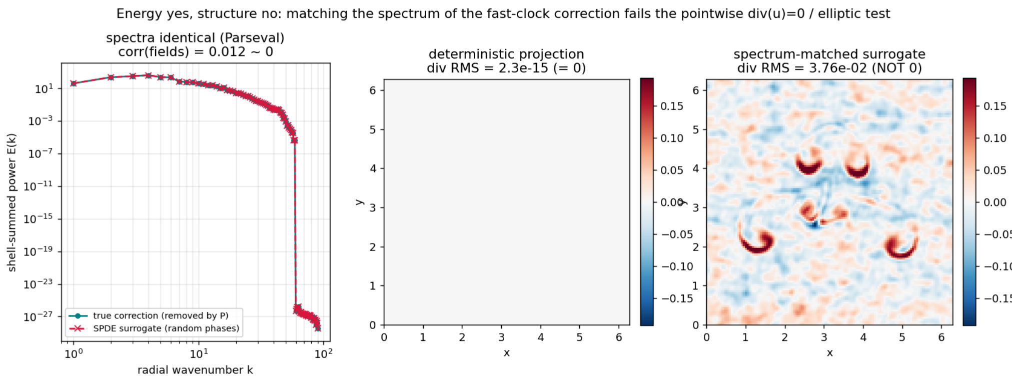

Figure 10: Spectrum-matched surrogate test. A phase-randomized field with the identical power spectrum as the true projection correction (Parseval-exact) fails to reproduce the correction pointwise: ensemble-median correlation $|r| = 0.07$ ($n = 100$), and the divergence constraint is left unrepaired (RMS $\nabla \cdot \mathbf{u} = 3.7 \times 10^{-2}$, compared to 2×10^{-15} for the true projection).

Table 4: Ratio of divergent to rotational kinetic energy (daily mean) from NCEP/NCAR Reanalysis 1 (R1) and NCEP–DOE Reanalysis 2 (R2). The minimum at 500 hPa corresponds to the classical level of non-divergence.

Level	R1: $KE_{\rm div}/KE_{\rm rot}$	R2 cross-check
850 hPa	4.0%	4.5%
500 hPa	0.7%	0.8%
250 hPa	1.0%	1.1%

not of one assimilation system.

Fourth, time-averaging acts as a low-pass filter on the fast clock: the six-hourly ratio at 500 hPa (1.1%) exceeds the daily-mean ratio (0.7%). Averaging scrubs the divergent component, exactly as acoustic averaging recovered the elliptic pressure in Section 5.2.

7 Discussion

7.1 Summary of measurements

Six independent measurements, spanning field observation, direct numerical simulation, controlled solvers, and reanalysis, consistently reveal the same structure: pressure and temperature respond to the same forcing through different operators with different timescales and different spatial footprints. The temporal signatures differ (local diurnal versus planetary semidiurnal); the spatial signatures differ (filamentary versus smooth, with a $1/k^4$ spectral gap); the dynamical demonstrations show the mechanism (hyperbolic-to-elliptic crossover, M^2 energy scaling, projection as synthesis); and the real atmosphere exhibits the expected energy partition with the correct vertical structure.

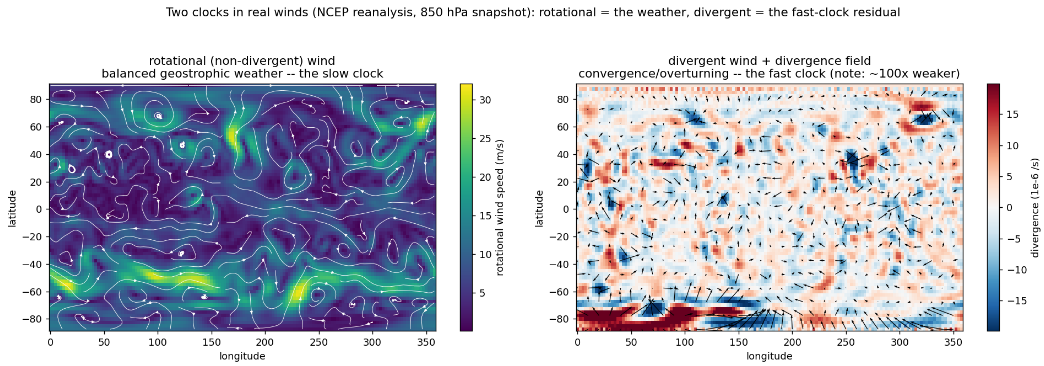

Figure 11: NCEP/NCAR winds, one analysis day. Left: rotational wind (synoptic highs, lows, and jets). Right: divergent wind (weak, confined to convergence and overturning zones).

7.2 Implications for turbulence closure

The measurements presented here have a direct implication for single-diffusivity turbulence closures. A closure that models both momentum transport (shaped by the global pressure field) and heat transport (shaped by local buoyant plumes) with a single eddy diffusivity and a fixed turbulent Prandtl number assumes that the two transport processes respond identically to changes in the shear-to-buoyancy ratio. The field data show otherwise: binned by Monin–Obukhov stability, the ratio of momentum to heat transport efficiency varies by a factor of 4.3 in winter and 6.0 in summer at the NEON WREF site (Appendix A). This variation is consistent with the two-operator structure: momentum transport is mediated by the global pressure field (fast clock), while heat transport is mediated by local buoyant plumes (slow clock), and their relative importance shifts with stability.

The spectrum-matched surrogate test of Section 5.3 further constrains the class of admissible closure corrections: stochastic augmentations that match the energy spectrum of the fast-clock correction but not its elliptic geometry cannot repair the divergence constraint. A constructive closure that addresses this limitation is developed in the companion paper (Saifelden 2026).

7.3 Scope and limitations

All numerical demonstrations are two-dimensional and periodic. No three-dimensional regularity claim is made. The reanalysis diagnostics are limited to spherical-harmonic degree $l \leq 35$ and are based on model-assimilated products, not raw observations. The field measurements span two months at the seasonal extremes (NEON) and one season at eight sites (ASOS); they establish the fingerprint of the operator split but do not constitute a global climatology.

7.4 Reproducibility

All figures and numerical results regenerate from the public repository at `https://github.com/abdh0saifelden2-debug/K`. Each analysis is a single command with fixed seeds, runnable on CPU. Input datasets are public: NEON DP4.00200.001 (data.neonscience.org), ASOS one-minute data via the Iowa Environmental Mesonet (mesonet.agron.iastate.edu), The Well Rayleigh–Bénard dataset (polymathic-ai), and NCEP/NCAR Reanalysis 1 via NOAA PSL OPeNDAP (psl.noaa.gov). See

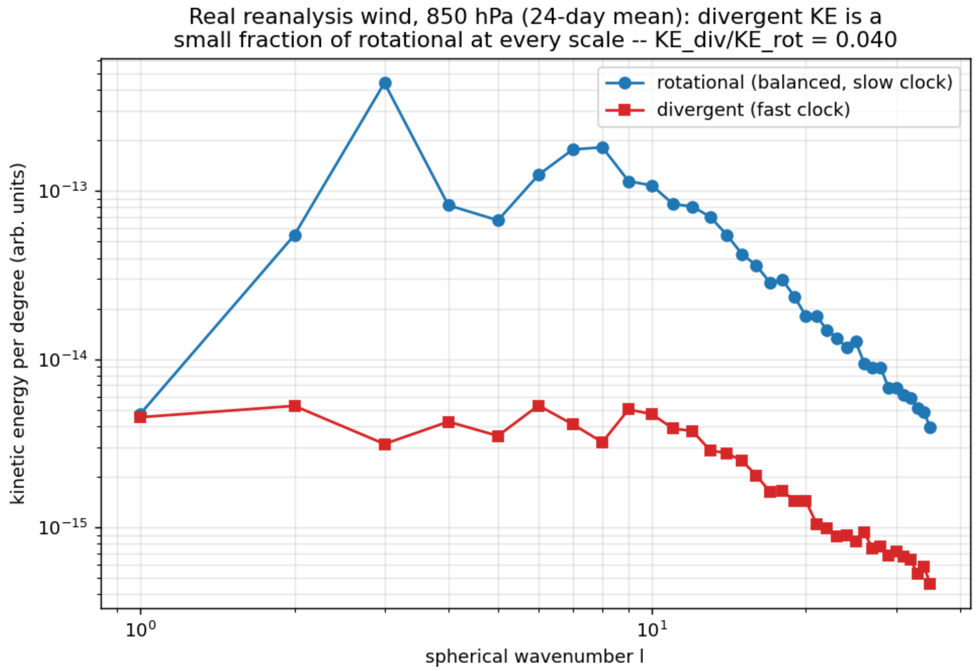

Figure 12: Kinetic-energy spectra by spherical-harmonic degree l: the divergent branch sits one to two decades below the rotational branch at every resolved scale.

Appendix A for the full command table.

Acknowledgments

[To be added.]

A Reproducibility

Table 5 lists the commands that regenerate each result in this paper.

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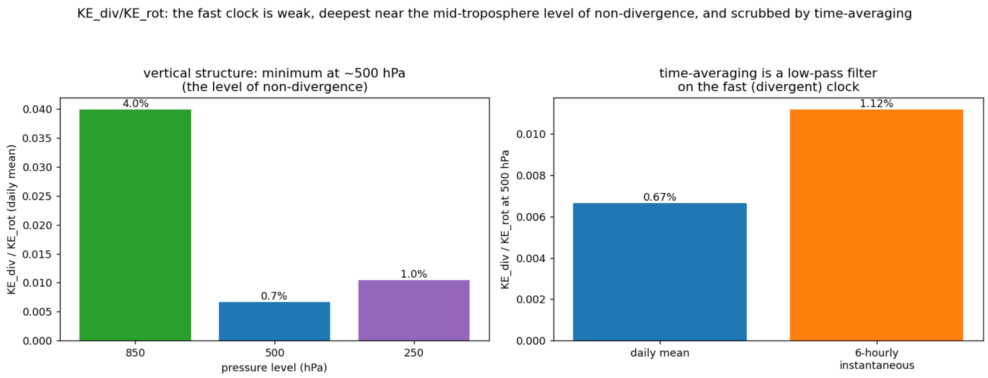

Figure 13: Vertical structure of $KE_{\mathrm{div}}/KE_{\mathrm{rot}}$: minimum at the level of non-divergence (∼500 hPa). Six-hourly values (open symbols) exceed daily-mean values (filled symbols), demonstrating that time-averaging acts as a low-pass filter on the fast clock.

Table 5: Reproducibility commands. All are runnable from the repository root with fixed seeds and CPU-only execution.

Result	Command
§3.1 NEON diurnal composites	`python general_two_clocks/run_analysis.py`
§3.1 Seasonal cross-check	`python general_two_clocks/run_neon_compare.py`
§3.2 ASOS latitude sweep	`python general_two_clocks/asos/fetch_iem.py && python general_two_clocks/run_asos_ci.py`
§4 RB DNS split	`python general_two_clocks/run_rb.py`
§4 *Ra* sweep	`python general_two_clocks/run_rb_ra_sweep.py`
§4 Green's function	`python general_two_clocks/run_elliptic_demo.py`
§7 Stability bins (E3)	`python general_two_clocks/run_stability.py`
§5.1 Linear crossover	`python general_two_clocks/run_compressible.py`
§5.2 Nonlinear Mach sweep	`python general_two_clocks/run_compressible_ns.py`
§5.3 Projection / surrogate	`python general_two_clocks/run_boussinesq.py`
§6 Reanalysis split	`python general_two_clocks/run_reanalysis.py`

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